SmartTriage System

Module-by-Module Deep Specification

Version 2.0

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Complete Technical Specification for Modules 1-27

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1 Module 1: Entry & Registration Engine

1.1 Purpose

To capture the legal and clinical start of care. Establishes identity, entry type, and begins the medico-legal clock.

1.2 Required Data Fields

- Arrival Time (auto, immutable)
- Entry Type (Ambulance / Walk-in / Referral)
- Referral Facility (if applicable)
- Referral Notes Upload
- Patient Name
- DOB
- Sex
- National ID / Hospital ID
- Contact Person
- Triage Nurse Assigned
- Mode of arrival (stretcher / walking / wheelchair)

1.3 Decision Rules

- Arrival Time auto-generated.
- Cannot proceed to triage without assigning triage clinician.
- If referral → referral document upload mandatory.

1.4 Alerts

- Duplicate patient ID warning
- Missing referral documentation
- Pediatric without guardian

1.5 Override Conditions

- Only admin role can edit demographic errors.
- Arrival time cannot be edited.

1.6 Timestamps

- Arrival time
- Registration completion time

1.7 Audit Logs

- Who registered patient
- Any demographic edits

1.8 Scenario

Ambulance arrives with 65-year-old male unconscious. Arrival time recorded: 10:03:12. Entry type: Ambulance. System flags "High acuity entry."

2 Module 2: mSAT Triage Engine

2.1 Purpose

To enforce Rwanda's modified South African Triage Scale objectively.

2.2 Required Data Fields

- Emergency signs checklist
- RR
- Pulse
- SBP
- Temp
- SpO
- AVPU
- Mobility
- Trauma (Y/N)
- Age
- Weight (peds mandatory)
- Triage Finish Time (auto)
- Category (system-generated only)

2.3 Decision Rules

- IF Emergency Sign = YES → RED
- ELSE IF SpO < 92% → RED
- ELSE IF TEWS 7 → RED
- ELSE IF TEWS 5–6 → ORANGE
- ELSE → Evaluate Very Urgent
- Medical + Trauma lists determine Very Urgent/Urgent status

2.4 Alerts

- Missing vital signs
- Abnormal values outside physiologic limits
- SpO < 92%
- TEWS rising

2.5 Override Conditions

- Senior physician only. Mandatory reason logged.

2.6 Timestamps

- Triage start
- Triage finish
- Category assigned
- Doctor notified
- Doctor attended

2.7 Audit Logs

- TEWS calculation record
- Overrides
- Escalations

2.8 Scenario

Patient walks in with chest pain. Vitals auto-captured. TEWS = 6. System assigns ORANGE. 10-minute countdown begins.

3 Module 3: TEWS Calculator Engine

3.1 Purpose

Automated, error-free scoring using mSAT thresholds.

3.2 Data Fields

- RR score
- Pulse score
- SBP score
- Temp score
- AVPU score
- Mobility score
- Trauma score
- Total TEWS (auto)

3.3 Rules

- Scoring table based on Rwanda mSAT parameters.
- Age-adjusted for pediatrics.

3.4 Alerts

- TEWS 7
- Rapid increase 2 in 10 min

3.5 Override Conditions

- Cannot manually edit TEWS score.

3.6 Scenario

Patient stable at TEWS 2. RR increases from 18 to 32. TEWS recalculates to 6. System upgrades to ORANGE.

4 Module 4: IoT Vital Signs Integration

4.1 Purpose

Eliminate manual entry errors. Provide continuous monitoring.

4.2 Data Fields

Auto-stream:

- RR
- Pulse
- SBP
- MAP
- SpO
- Temp
- ECG (if connected)

4.3 Rules

- Every 2 min rolling average.
- Data validation against physiologic limits.

4.4 Alerts

- SpO $< 92\%$
- SBP < 90
- RR > 30 or < 8
- HR > 130 or < 40

4.5 Override Conditions

- Cannot disable monitoring unless clinician signs reason.

4.6 Scenario

SpO drops gradually $95 \rightarrow 90 \rightarrow 88$. System triggers RED alert before collapse.

5 Module 5: Dynamic Re-triage AI Engine

5.1 Purpose

Predict deterioration before arrest.

5.2 Data Inputs

- Continuous vitals
- TEWS trend
- Lab results
- Medical history

5.3 Rules

- Trend slope analysis
- Variability detection
- Escalate if deterioration pattern identified

5.4 Alerts

- Category upgrade
- Immediate staff notification
- Alarm escalation tiers

5.5 Override Conditions

- Downgrade requires senior review.

5.6 Scenario

YELLOW patient waiting. RR increasing. Pulse rising. System upgrades to ORANGE before crash.

6 Module 6: Pediatric Logic Engine

6.1 Purpose

To ensure all triage, scoring, medication dosing, and alerts are age-adjusted and pediatric-safe according to Rwanda MoH and WHO pediatric emergency standards. Children are not small adults. Vital ranges differ. Drug dosing differs. Deterioration patterns differ. This module prevents fatal pediatric misclassification.

6.2 Required Data Fields

Mandatory if Age < 15:

- Exact Age (in months if <2 years)
- Weight (kg) — Mandatory

- Height (optional but recommended)
- Immunization status
- Birth history (if <1 year)
- Developmental baseline
- Pediatric AVPU
- Caregiver name
- Guardian contact
- Vital signs must use pediatric reference ranges:
  - Age-adjusted RR
  - Age-adjusted HR
  - Age-adjusted BP

6.3 Decision Rules

- TEWS uses pediatric thresholds.
- Weight required before medication ordering.
- Drug dose auto-calculated (mg/kg).
- If entered dose exceeds safe range → hard stop.
- SpO $< 94\%$ in infants triggers higher alert sensitivity.

6.4 Alert Thresholds

- HR above age-appropriate tachycardia range
- RR above age-appropriate distress range
- Weight missing
- Drug dose >10% above recommended mg/kg
- Fever in infant <3 months → ORANGE minimum

6.5 Override Conditions

Pediatric dose override requires:

- Senior physician role
- Justification
- Double signature

6.6 Required Timestamps

- Weight entry time
- Dose calculation time
- Pediatric alert acknowledgement time

6.7 Audit Logs

- Weight edits
- Dose overrides
- Pediatric category upgrades/downgrades

6.8 Scenario

8-month-old baby with fever and lethargy. Vitals: HR 180, RR 50, Temp 39.5. Adult system might under-triage. Pediatric Engine: Recognizes age-specific tachycardia. Flags high risk. Auto-assigns ORANGE minimum. Prompts sepsis screening. Life potentially saved.

7 Module 7: Sepsis Detection Engine

7.1 Purpose

Early identification of sepsis and septic shock, aligned with WHO and Rwanda MoH emergency guidelines. Sepsis is time-sensitive. Mortality rises every hour antibiotics are delayed.

7.2 Required Data Fields

- Temperature
- HR
- RR
- SBP
- SpO
- Lactate (if available)
- WBC (if available)
- Suspected infection source
- Urine output (ICU)
- Mental status

7.3 Decision Rules

Sepsis Alert triggers if:

- Suspected infection + 2 of:
  - RR > 22
  - SBP < 100
  - Altered mentation
  - Temp > 38.5 or < 36
  - HR > 100
- Septic Shock: Persistent hypotension, Lactate elevated (if available)
- When triggered: Category auto ORANGE minimum, Sepsis bundle timer (1-hour clock), Antibiotic recommendation prompt

7.4 Alerts

- Sepsis suspected
- Septic shock risk
- 1-hour antibiotic deadline approaching
- Hypotension persistent >5 minutes

7.5 Override Conditions

Cannot dismiss sepsis alert without:

- Entering alternative diagnosis
- Senior review

7.6 Required Timestamps

- Sepsis detection time
- Antibiotic ordered time
- Antibiotic administered time
- Fluid bolus start time

7.7 Audit Logs

- Time to antibiotics
- Time to fluids
- Alert ignored/delayed

7.8 Scenario

45-year-old with fever and confusion. Vitals: RR 28, SBP 95, Temp 39. System triggers Sepsis Alert. 1-hour clock starts. Antibiotics administered at 32 minutes. Mortality risk reduced.

8 Module 8: Stroke / MI Fast-Track Module

8.1 Purpose

To enforce time-critical pathways for stroke and myocardial infarction. Time = brain. Time = myocardium.

8.2 Required Data Fields

For Stroke:

- Time of symptom onset
- FAST assessment
- GCS
- Blood glucose
- BP

For MI:

- Chest pain onset time
- ECG time
- Troponin ordered time
- Risk factors
- Pain severity

8.3 Decision Rules

- If FAST positive: Auto ORANGE minimum, Door-to-CT timer, Neurology notification
- If chest pain + ECG abnormal: STEMI alert, Door-to-ECG 10 min enforcement, Cardiology notification

8.4 Alerts

- CT delay >20 minutes
- ECG not performed within 10 minutes
- Thrombolysis window expiring

8.5 Override Conditions

Cannot close stroke pathway without:

- CT result OR
- Reason documented

8.6 Required Timestamps

- Arrival
- ECG performed
- CT ordered
- CT completed
- Thrombolysis time

8.7 Audit Logs

- Door-to-ECG time
- Door-to-CT time
- Door-to-needle time

8.8 Scenario

62-year-old sudden facial droop. Arrival 14:02. FAST positive. CT ordered 14:04. CT done 14:16. Timer enforced rapid response.

9 Module 9: Hypoglycemia Enforcement Module

9.1 Purpose

Prevent missed hypoglycemia in altered patients. Hypoglycemia is reversible but fatal if missed.

9.2 Required Data Fields

Mandatory glucose entry if:

- Altered mental status
- Seizure
- Diabetic
- Stroke symptoms
- Pediatric lethargy

Fields:

- Blood glucose value
- Time measured
- Method (glucometer/lab)

9.3 Decision Rules

- If glucose < 3.0 mmol/L: $\rightarrow$ RED, Immediate dextrose protocol
- If glucose 3.0–4.0: $\rightarrow$ ORANGE minimum
- Cannot complete assessment without glucose value if criteria met

9.4 Alerts

- Missing glucose
- Critically low glucose
- Recheck glucose after treatment

9.5 Override Conditions

- Cannot bypass glucose requirement.

9.6 Required Timestamps

- Glucose measurement time
- Dextrose administration time
- Recheck time

9.7 Audit Logs

- Missed glucose attempts
- Treatment delay

9.8 Scenario

Unresponsive diabetic arrives. Glucose measured: 1.9 mmol/L. RED assigned. Dextrose given within 2 minutes. Patient wakes.

10 Module 10: Infection Isolation & Public Health Module

10.1 Purpose

Prevent spread of infectious diseases in ED and ICU. Critical for Rwanda context: TB, Viral hemorrhagic fevers, Measles, COVID-like illnesses.

10.2 Required Data Fields

- Cough duration
- Travel history
- TB exposure
- Rash presence
- Bleeding symptoms
- Fever
- Isolation required (auto-flag)

10.3 Decision Rules

If:

- Cough >2 weeks + weight loss → TB screening
- Fever + bleeding → high-risk infection alert
- Rash + fever (child) → measles alert
- Respiratory symptoms → droplet precaution
- System assigns: Isolation flag, PPE requirement

10.4 Alerts

- Isolation not initiated
- PPE not confirmed
- Suspected outbreak case

10.5 Override Conditions

Isolation removal requires:

- Negative testing OR
- Senior physician documentation

10.6 Required Timestamps

- Isolation initiated time
- PPE confirmation time
- Public health notification time

10.7 Audit Logs

- Isolation breaches
- Alert acknowledgments
- Public health notifications sent

10.8 Scenario

Patient with cough 3 weeks, weight loss. System flags TB suspicion. Isolation required. Prevents exposure to other patients.

11 Module 11: Clinical Documentation System

11.1 Purpose

To ensure legally compliant, complete, structured, time-stamped clinical documentation. Documentation must meet Rwanda MoH standards, be medico-legally defensible, be structured for audit and analytics, and prevent incomplete notes.

11.2 Required Documentation Components

Each patient encounter must include:

- Chief Complaint
- History of Present Illness (structured)
- Past Medical History
- Medication History
- Allergies
- Physical Examination
- Vital Signs (auto-pulled)
- Assessment
- Plan
- Clinician Name & Role
- Digital Signature
- Timestamp (auto-generated)

11.3 System Logic

- No discharge without completed note.
- Vital signs auto-embedded (cannot manually rewrite).
- Medication plan must reference dose.
- Red-category patients require reassessment note.

11.4 Alerts

- Incomplete documentation
- Missing signature
- Discharge without assessment
- Copy-paste overuse detection
- Time gap $>$ 2 hours without update (RED/ORANGE)

11.5 Override Conditions

- Emergency documentation delay allowed for RED patients.
- Must complete note within 30 minutes post-stabilization.

11.6 Required Timestamps

- Note started
- Note completed
- Signature time
- Edits logged

11.7 Audit Logs

- Edit history
- Late documentation flags
- Discharge documentation compliance

11.8 Scenario

Patient discharged without medication instructions. System blocks discharge. Forces completion of prescription documentation. Prevents legal risk and medication errors.

12 Module 12: Medication Safety Engine

12.1 Purpose

Prevent: Wrong drug, Wrong dose, Wrong patient, Allergy reactions, Drug interactions. This module is life-critical.

12.2 Required Data Fields

- Drug name (standardized list)
- Dose
- Unit
- Route
- Frequency
- Duration
- Weight (if pediatric)
- Renal function (if available)
- Allergy list

12.3 Decision Rules

When medication is entered, system checks:

- Allergy match
- Dose range (adult or pediatric)
- Renal adjustment needed
- Drug-drug interaction
- Maximum daily dose
- Duplicate therapy
- If error: Hard stop for critical errors, Warning for moderate risk

12.4 Alerts

- Allergy conflict (Hard stop)
- Overdose risk
- Contraindicated drug
- High-risk medication (e.g., insulin, anticoagulants)
- Pediatric dose mismatch
- Renal toxicity risk

12.5 Override Conditions

Hard stop overrides require:

- Senior physician login
- Written justification
- Second verification (for high-risk meds)

12.6 Required Timestamps

- Medication ordered
- Pharmacist verification
- Medication administered

12.7 Audit Logs

- Override usage
- Adverse drug event reports
- Near-miss tracking

12.8 Scenario

Child weight 12kg. Paracetamol entered 1000mg. System detects overdose. Blocks order. Calculates safe dose automatically. Prevents toxicity.

13 Module 13: Laboratory Integration Module

13.1 Purpose

Integrate laboratory services into SmartTriage workflow. Ensure fast ordering, automatic result capture, critical value alerts, and no missed results.

13.2 Required Data Fields

- Test ordered
- Ordering clinician
- Order time
- Priority (STAT / Routine)
- Specimen collection time
- Result value
- Reference range
- Result time

13.3 Decision Rules

- Critical values trigger RED alert.
- STAT orders prioritized.
- Abnormal labs auto-highlighted.
- Sepsis labs link to Sepsis Engine.
- Troponin links to MI module.

13.4 Critical Value Examples

- K+ > 6.0
- Glucose < 2.5
- Lactate > 4
- Hemoglobin < 6
- Positive malaria smear (if severe symptoms)

13.5 Override Conditions

- Critical result must be acknowledged. Cannot be ignored without documentation.

13.6 Required Timestamps

- Order time
- Specimen collected
- Result available
- Clinician acknowledged

13.7 Audit Logs

- Time to acknowledgment
- Missed critical result
- Turnaround times

13.8 Scenario

Lab returns potassium 6.5. System auto-alerts RED. Physician notified. Calcium gluconate ordered immediately. Cardiac arrest prevented.

14 Module 14: Radiology Integration Module

14.1 Purpose

Streamline imaging workflow and prevent delayed interpretation.

14.2 Required Data Fields

- Imaging ordered
- Indication
- Priority
- Order time
- Completion time
- Report time
- Radiologist name
- Preliminary vs final report

14.3 Decision Rules

- Stroke CT auto-priority.
- Trauma CT auto-priority.
- Delayed imaging $>$ target $\rightarrow$ alert.
- Critical findings $\rightarrow$ RED alert.

14.4 Alerts

- Delayed CT (>20 min stroke)
- Pneumothorax
- Intracranial bleed
- Free abdominal air
- Massive PE

14.5 Override Conditions

- Imaging cannot be closed without: Final report OR Cancellation reason.

14.6 Required Timestamps

- Order time
- Imaging start
- Imaging complete
- Report available
- Report acknowledged

14.7 Audit Logs

- Door-to-CT metrics
- Critical result communication delay

14.8 Scenario

Stroke patient CT delayed 30 minutes. System alerts imaging department. Escalation triggered. Door-to-needle time protected.

15 Module 15: Clinical Pathway Automation Engine

15.1 Purpose

Standardize care using automated evidence-based pathways. This module reduces variation, errors, and missed steps.

15.2 Supported Pathways (Initial Version)

- Sepsis Bundle
- Stroke Pathway
- STEMI Pathway
- Pediatric Fever
- Trauma Primary Survey
- Severe Malaria
- Obstetric Emergency
- Diabetic Ketoacidosis

15.3 Decision Logic

When condition detected, system:

- Suggests pathway
- Pre-populates order sets
- Starts timer
- Displays checklist

15.4 Alerts

- Step incomplete
- Delay in intervention
- Window expiring
- Checklist not completed

15.5 Override Conditions

Pathway dismissal requires:

- Clinical reason
- Senior physician role (for RED)

15.6 Required Timestamps

- Pathway triggered
- Checklist completion
- Bundle completion

15.7 Audit Logs

- Bundle compliance rate
- Step completion time
- Outcome tracking

16 Module 16: ICU Escalation Logic

16.1 Purpose

Identify patients requiring ICU admission early and enforce escalation to critical care. Delays in ICU transfer increase mortality.

16.2 Required Data Inputs

- Current triage category
- Vital signs trends (not single reading)
- Oxygen requirement
- Vasopressor use
- GCS score
- Lactate (if available)
- Organ failure indicators
- Sepsis status
- Mechanical ventilation status

16.3 Decision Logic

ICU Alert triggers if any of:

- RED category persistent >10 minutes
- Need for mechanical ventilation
- Vasopressor initiated
- GCS 8
- Lactate 4
- Refractory hypotension
- Multi-organ failure criteria met

System actions: Auto ICU consult notification, Bed availability check, Escalation timer starts, Senior physician alert.

16.4 Alerts

- ICU criteria met
- ICU consult not acknowledged
- No ICU bed assigned within defined threshold
- Recurrent deterioration

16.5 Override Conditions

ICU escalation cannot be dismissed without:

- Documented clinical justification
- Senior-level approval
- Re-evaluation scheduled within 30 minutes

16.6 Required Timestamps

- ICU criteria met
- ICU notified
- ICU accepted
- Transfer initiated
- Transfer completed

16.7 Audit Logs

- Time from ICU trigger to transfer
- ICU denial reasons
- Mortality correlation tracking

16.8 Scenario

Septic patient requires norepinephrine. System detects vasopressor use. ICU alert triggered. Transfer completed within 25 minutes. Delay avoided.

17 Module 17: Referral & Inter-Hospital Transfer Module

17.1 Purpose

Standardize safe patient referral between facilities. Critical for district-to-tertiary transfers.

17.2 Required Data Fields

- Referring facility
- Receiving facility
- Diagnosis
- Stability status

- Vital signs
- Reason for transfer
- Escort personnel
- Transport method
- Medications administered
- Imaging/lab summary

17.3 Decision Logic

System requires:

- Stabilization checklist completed
- Vital signs documented within 10 minutes of departure
- Referral summary auto-generated
- Receiving facility confirmation
- High-risk transfer flags: Intubated patient, Hemodynamically unstable, Neonatal transfer, Obstetric emergency

17.4 Alerts

- Transfer initiated without stabilization
- Missing referral documentation
- No receiving confirmation
- Ambulance delay

17.5 Override Conditions

Transfer cannot proceed without:

- Stabilization checklist completion OR
- Senior override with justification

17.6 Required Timestamps

- Transfer decision
- Referral sent
- Receiving accepted
- Departure time
- Arrival time

17.7 Audit Logs

- Transfer delays
- Stabilization compliance
- Mortality during transfer

17.8 Scenario

District hospital identifies stroke beyond thrombolysis window. Referral initiated to tertiary center. Summary auto-sent. Receiving confirms. Transfer documented end-to-end. No paperwork lost.

18 Module 18: Quality Metrics & Dashboard Engine

18.1 Purpose

Provide real-time hospital performance monitoring. This transforms SmartTriage into a quality improvement tool.

18.2 Core Metrics Tracked

- Door-to-ECG time
- Door-to-CT time
- Door-to-antibiotic time
- Sepsis bundle compliance
- ICU transfer time
- Mortality by triage category
- Left without being seen
- Average ED length of stay
- Documentation completion rate
- Medication override rate

18.3 System Features

- Real-time dashboard
- Department-level filtering
- Monthly performance reports
- Benchmarking across hospitals
- Trend visualization

18.4 Alerts

- KPI below threshold
- Mortality spike
- Recurrent delay pattern
- Excess override usage

18.5 Governance Controls

- Only admin roles access hospital-wide data
- National authority view aggregated data only

18.6 Audit Logs

- Report generation
- Data export history
- Dashboard access tracking

18.7 Impact

ED identifies: Door-to-antibiotic average 82 minutes. Improvement plan implemented. Average reduced to 48 minutes. Measurable mortality improvement.

19 Module 19: AI-Assisted Risk Prediction Engine

19.1 Purpose

Use machine learning to predict clinical deterioration before it is visible. Not to replace clinicians — but to augment them.

19.2 Data Sources

- Vital sign trends
- Lab trends
- Demographics
- Diagnosis
- Medication use
- Comorbidities
- Triage history

19.3 Prediction Capabilities

- 12-hour deterioration risk
- 24-hour ICU need prediction
- Sepsis progression risk
- Cardiac arrest early warning
- Readmission risk

19.4 Alerts

- High predicted deterioration risk
- Escalation recommended
- Reassessment suggested
- Alert must show: Risk score, Confidence level, Contributing factors

19.5 Override Conditions

- AI alerts can be dismissed but require clinical reasoning entry.

19.6 Required Timestamps

- Prediction generated
- Alert acknowledged
- Action taken

19.7 Audit Logs

- AI alert acceptance rate
- False positive tracking
- Outcome validation

19.8 Scenario

Patient stable initially. AI detects rising RR + subtle BP drop. Flags high risk of sepsis progression. Reassessment performed. Patient transferred early. Crisis prevented.

20 Module 20: National Health Data Reporting Interface

20.1 Purpose

Enable secure data reporting to Ministry of Health and public health authorities. Supports disease surveillance, policy planning, resource allocation, and outbreak detection.

20.2 Data Categories Reported

Aggregated only (no patient identifiers unless legally required):

- Triage volumes
- Sepsis cases
- Stroke cases
- MI cases
- Maternal emergencies
- Pediatric mortality
- TB suspects
- Isolation cases
- ICU utilization
- Antibiotic timing compliance

20.3 Data Governance

- De-identified by default
- Encrypted transmission
- Role-based access
- Consent logic compliance
- Configurable reporting intervals

20.4 Alerts

- Sudden infectious disease spike
- Mortality cluster
- Unusual triage pattern

20.5 Override Conditions

- National-level reporting cannot be disabled without: Administrative authority, Documented legal reason.

20.6 Required Timestamps

- Data exported
- Data received
- Report generated

20.7 Audit Logs

- All data transmissions
- Access logs
- External reporting logs

20.8 Impact Scenario

Spike in febrile rash cases detected. System flags potential outbreak. Ministry notified within 24 hours. Early containment possible.

21 Module 21: System Resilience & Offline Continuity Engine

21.1 Purpose

Ensure SmartTriage functions during power outages, internet disruption, server failure, cyber incidents, and disaster conditions. This is critical for district hospitals and rural facilities.

21.2 Core Functional Requirements

- Local device caching of active patients
- Offline triage mode
- Deferred sync logic
- Conflict resolution protocol
- Local encrypted data storage
- Automatic reconnection & sync

21.3 Operational Logic

If connectivity lost:

- System switches to Offline Mode
- All entries timestamped locally
- Critical alerts continue locally
- Data queued for secure sync

Upon reconnection:

- Automatic encrypted synchronization
- Conflict reconciliation (latest timestamp logic)
- Audit reconciliation

21.4 Alerts

- Offline mode activated
- Sync conflict detected
- Data integrity mismatch
- Battery critical warning (tablet mode)

21.5 Override Conditions

- Offline deletion of patient records is prohibited.

21.6 Required Timestamps

- Offline mode start
- Offline mode end
- Sync start
- Sync completion

21.7 Audit Logs

- Offline duration per facility
- Sync failures
- Data reconciliation events

21.8 Scenario

District hospital loses internet for 3 hours. Triage continues uninterrupted. Sepsis alerts still function. System syncs when connection restored. No data lost.

22 Module 22: Cybersecurity & Threat Monitoring Engine

22.1 Purpose

Protect patient data and prevent system compromise. Healthcare systems are high-value cyber targets.

22.2 Security Layers

- End-to-end encryption
- Multi-factor authentication
- Role-based access
- Session timeout
- IP monitoring
- Device binding
- Intrusion detection system
- Automated lockout after failed attempts

22.3 Detection Logic

System monitors:

- Repeated login failures
- Unusual data export volume
- Access outside shift hours
- Admin privilege misuse
- Abnormal activity spikes

22.4 Alerts

- Suspicious login
- Data breach attempt
- Unauthorized export
- Privilege escalation attempt

22.5 Response Protocol

- Immediate account lock
- Admin notification
- National authority notification (if major breach)
- Forensic audit log preservation

22.6 Required Timestamps

- Login attempt
- Lockout time
- Incident detection
- Resolution time

22.7 Audit Logs

- Login history
- Role changes
- Data exports
- Security incidents

22.8 Impact

Prevents patient data theft, system manipulation, and ransomware exploitation.

23 Module 23: Clinical Governance & Policy Control Engine

23.1 Purpose

Ensure SmartTriage remains aligned with Rwanda MoH policy updates, WHO guideline revisions, hospital protocol changes, and antibiotic stewardship policies.

23.2 Governance Features

- Centralized policy update control
- Version control system
- Change approval workflow
- Effective date scheduling
- Rollback capability

23.3 Change Control Logic

Any update to:

- Triage thresholds
- Sepsis criteria
- Drug dose ranges
- Pathway steps

Requires:

- Proposal submission
- Clinical governance review
- Approval vote
- Version release
- Deployment notification

23.4 Alerts

- Outdated guideline in use
- Policy conflict detected
- Unapproved modification attempt

23.5 Override Conditions

Emergency policy update allowed only for:

- Outbreak
- New national directive
- Must be formally documented within 24 hours.

23.6 Required Timestamps

- Policy submitted
- Approved
- Activated
- Deactivated

23.7 Audit Logs

- Version history
- Policy authors
- Change justification
- Rollback events

23.8 Scenario

MoH updates malaria treatment protocol. SmartTriage pushes update nationwide. All facilities aligned same day. No variation in dosing.

24 Module 24: Patient Safety Incident Reporting System

24.1 Purpose

Capture, analyze, and reduce adverse events. Promotes a culture of safety, not blame.

24.2 Reportable Events

- Medication error
- Delayed intervention
- Near-miss
- Equipment failure
- Documentation lapse
- Unexpected deterioration
- Death review trigger

24.3 System Logic

Any user can submit:

- Anonymous or named report
- Event category
- Description
- Contributing factors
- Harm severity level

System auto-links:

- Relevant timestamps
- Clinical logs
- Medication data

24.4 Alerts

- High-severity incident
- Recurrent similar incidents
- Department trend spike

24.5 Investigation Workflow

- Incident submitted
- Safety officer notified
- Root cause analysis initiated
- Action plan logged
- Follow-up review scheduled

24.6 Required Timestamps

- Incident reported
- Review started
- Review completed
- Corrective action implemented

24.7 Audit Logs

- Incident rate per 1,000 patients
- Repeat patterns
- Time to resolution

24.8 Impact

Medication near-miss detected. Pattern analysis shows dosing confusion. UI redesigned. Error rate reduced 60%.

25 Module 25: Workforce Performance & Training Analytics Engine

25.1 Purpose

Support continuous clinical skill improvement. Not punitive — developmental.

25.2 Metrics Tracked

- Triage accuracy vs outcome
- Time to intervention
- Documentation completion
- Alert acknowledgment time
- Medication override frequency
- Pathway compliance rate

25.3 System Features

- Individual performance dashboard (private)
- Departmental benchmarking
- Training recommendation engine
- Certification tracking
- Competency alerts

25.4 Alerts

- Repeated missed sepsis alerts
- High documentation delays
- Recurrent dosing overrides
- Skill gap identified

25.5 Governance Controls

- Individual data visible only to: Clinician, Supervisor, Governance body (if escalated)
- National view only aggregated data.

25.6 Required Timestamps

- Training assigned
- Training completed
- Competency validated

25.7 Audit Logs

- Certification status
- Refresher training history
- Skill improvement trends

25.8 Impact

System detects multiple nurses missing early sepsis alerts. Targeted training deployed. Compliance increases. Mortality decreases.

26 Module 26: Longitudinal Medical History Engine

26.1 Purpose

Maintain structured, lifetime patient clinical history across encounters and facilities. This is not a note. This is structured clinical memory.

26.2 Core Components

- Chronic conditions
- Surgical history
- Allergy history
- Vaccination history
- Obstetric history
- Previous ICU admissions
- Prior sepsis episodes
- Medication history (chronic)
- Baseline vital trends
- Baseline lab trends
- Disability status
- Advanced directives

26.3 Design Principles

- Structured data (not just free text)
- Version-controlled
- Editable only by authorized roles
- Every change logged
- Historical version preserved

26.4 Interaction With Other Modules

- Triage pulls chronic conditions
- Medication engine checks chronic renal disease
- AI engine uses history trends
- Referral auto-includes summary
- ICU module sees baseline status

26.5 Alerts

- High-risk comorbidity detected
- Allergy mismatch
- Missing critical chronic diagnosis

26.6 Audit Logs

- History edits
- Access logs
- Conflict resolution

26.7 Example

Patient arrives hypotensive. History shows: Chronic heart failure, CKD stage 4. System modifies fluid recommendations. Safer management.

27 Module 27: Health Information Exchange (HIE) & Interoperability Engine

27.1 Purpose

Allow secure data exchange between district hospitals, referral hospitals, national registries, laboratory networks, and insurance systems.

27.2 Standards Compliance

Should support:

- HL7/FHIR structured data
- Encrypted APIs
- National ID integration
- Facility identifier mapping

27.3 Capabilities

- Query external records
- Push referral data automatically
- Sync lab results from central labs
- Import vaccination history
- Insurance eligibility verification

27.4 Alerts

- Duplicate patient detected
- ID mismatch
- Data integrity conflict

27.5 Governance

- Consent-based data exchange
- Role-restricted external queries
- Full audit trail

27.6 Impact

Patient transferred from rural hospital. Receiving hospital instantly retrieves prior CT, previous labs, and medication list. No duplication. No delay. No risk from missing data.